CH 2 — Linear Transformations

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2.1 Linear Transformations Between Abstract Vectors

📘 **Definition — Linear Map**

Let V, W be vector spaces over $\mathbb{F}$

A **linear map** from V to W is a function $L : V \to W$ satisfiying

$$L(\vec{x} + \vec{y}) = L(\vec{x}) + L(\vec{y})$$
$$L(c\vec{x}) = cL(\vec{x})$$

Remarks:

1. L is linear if and only if $L(c\vec{x} + \vec{y}) = cL(\vec{x}) + L(\vec{y}), \quad \forall \vec{x}, \vec{y} \in V, c \in \mathbb{F}$
2. $L\left(\vec{0}\right) = \vec{0}$

Proof

$L\left(\vec{0}\right) = L\left(\vec{0} + \vec{0}\right) = L\left(\vec{0}\right) + L\left(\vec{0}\right) \implies \vec{0} = L\left(\vec{0}\right)$

□

Examples:

1. Zero map: $\vec{z} : V \to W$ defined by $\vec{z}(x) = \vec{0}_W, \quad \forall \vec{x} \in V$
2. Identity map: $\mathrm{Id} : V \to W$ defined by $\vec{x} \to \vec{x}$
3. B-coordinates. $[]_B : V \to \mathbb{F}^n$ is linear defined by $\vec{x} \to [\vec{x}]_B$
4. Let $A \in M_{m\times n}(\mathbb{F})$, let $L_A : \mathbb{F}^n \to \mathbb{F}^m$ defined by $\vec{x} \to A\vec{x}$
5. Evaulation is linear. Let $t_0 \in \mathbb{F}$, $\mathrm{ev}_0 : P_n(\mathbb{F}) \to \mathbb{F}$ defined by $p(x) \to p(t_0)$ $\mathrm{ev}_0 : P_2(\mathbb{F}) \to \mathbb{F} \implies p(x) \to p(0) \implies a + bx + cx^2 \to a)$
6. Differentiation and integration are linear.

   Consider $D : P_n(\mathbb{F}) \to P_{n-1}(\mathbb{F})$ defined by $p(x) \to p'(x)$

   This is linear because

   $$D(p + q) = (p + q)' = p' + q' = D(p) + D(q)$$
   $$D(cp) = cp' = cD(p)$$

   Similarly, $L : P_{n(\mathbb{R})} \to \mathbb{R}$ defined by $p(x) \to \int_0^1 p(x)\,\mathrm{d}x$
7. Trace map is linear. $M_{n\times n}(\mathbb{F}) : A \to \mathrm{tr}(A)$

💡 **Tip — Combining Linear Maps**
Suppose $L, T : V \to W$.

1. $L + T$ is linear
2. cL is linear $\forall c \in \mathbb{F}$
3. If $S : W \to U$ is linear, then $S \circ T : V \to W$ is linear

Proof

If $S : W \to U$ is linear, then $S \circ T : V \to W$ is linear

$$\begin{aligned}(S \circ T)(c\vec{x} + \vec{y}) &= S(T(c(\vec{x}) + \vec{y})) \\ &= S(cT(\vec{x}) + T(\vec{y})) = cS(T(\vec{x})) + S(T(\vec{y})) \\ &= c(S \circ T)(\vec{x}) + (S \circ T)(\vec{y})\end{aligned}$$

□

💡 **Tip — Non-Linear Maps**(Rule of thumb)

1. Anything with $L\left(\vec{0}\right) \neq \vec{0}$ is probably non-linear
2. Anything involving muliplicators scaling probably is non-linear

Examples of non-linear maps:

1. $L(p(x)) = p(x) + 1$ is non-linear
2. $L(p(x)) = [p(x)]^2$ is non-linear
3. $L(A) = L(\det A)$ is non-linear
4. $L(p(x)) = xp(x)$ IS LINEAR

2.2 Rank-Nullity

IMPORTANT:

1. State and Prove the Fundamental of Linear Algebra(Rank-Nullity Theorem)
2. Show linear maps $\iff$ matrices

📘 **Definition — Kernel and Range**

Let $L : V \to W$ be a linear map.

1. $\text{Ker}(L) = \{\vec{v} \in V : L\left(\vec{v} = \vec{0}_W\right\}$ is a subspace of V
2. $\text{Range}(L) = \{\vec{w} \in W : \vec{w} = L(\vec{v}))\}$ is a subspace of W

Proof

$\text{Ker}(L) = \{\vec{v} \in V : L\left(\vec{v} = \vec{0}_W\right\}$ is a subspace of V

$L\left(\vec{0}_V\right) = \vec{0}_W, \vec{0}_V \in \text{Ker}(L)$

Let $\vec{x}, \vec{y} \in \text{Ker}(L), c \in \mathbb{F}$

$L(c\vec{x} + \vec{y}) = cL(\vec{x}) + L(\vec{y}) = \vec{0}_W \in W$, thus $c\vec{x} + \vec{y} \in \text{Ker}(L)$

□

Example:

$L = []_B \implies L : V \to \mathbb{F}^n$

$\text{Ker}(L) = \left\{ \vec{x} \in V : [\vec{x}]_B = \begin{bmatrix} 0 \\ \dots \\ 0 \end{bmatrix} \right\} = \left\{\vec{0}_V\right\}$, by the Uniqueness of B-coordinates

$\text{Range}(L) = \{[\vec{x}]_B : \vec{x} \in V\} = \mathbb{F}^n$

Theorem — Nullity-Rank Theorem

The **nullity** of L is

$$\text{nullity } L = \dim(\text{Ker } L)$$

The **rank** of L is

$$\text{rank } L = \dim(\text{Range } L)$$

Let $L : V \to W$ be linear mapping.

$$\dim V = \text{nullity } L + \text{rank } L$$
$$\dim V - \text{nullity } L = \text{rank } L$$

Total degrees - lost degrees = output degrees

Examples:

1. Let $A \in M_{m \times n}(\mathbb{F}), L_A : \mathbb{F}^n \to \mathbb{F}^m, \vec{x} \to A\vec{x}$

   a. Ker $L_A = \text{Null} A$ is the nullspace

   b. Range $L_A = \text{Col } A$ is the column space

   Proof

   $$\begin{aligned} \text{Ker } L_A &= \{\vec{x} \in \mathbb{F}^n : L_A(\vec{x}) = 0\} \\ &= \left\{\vec{x} \in \mathbb{F}^n : A\vec{x} = \vec{0}\right\} \\ &= \text{Null } A \end{aligned}$$

$$\begin{aligned}\text{Range } L_A &= \{\vec{x} \in \mathbb{F}^n : \vec{y} = L_A(\vec{x})\} \\ &= \{\vec{x} \in \mathbb{F}^n : \vec{y} = 0A\vec{x}\} \\ &= \{y \in \mathbb{F}^m : \vec{y} = [\vec{a}_1, ..., \vec{a}_n]\vec{x}\} = \{y \in \mathbb{F}^m : \vec{y} = \vec{a}_1 x_1 + ...\vec{a}_n x_n, (x_1, ..., x_n) \in \mathbb{F}\} \\ &= \text{Span } \{\vec{a}_1, ..., \vec{a}_n\} = \text{Col } A\end{aligned}$$

□

This example will tell us eventually how to compute $\text{Ker}(L)$ and Range (L) in general

This result gives useful interpretation for the rank $A = \dim\ (\text{Range}(L))$

2. Let $D : P_n(\mathbb{R}) \to P_{n-1}(\mathbb{R}), p(x) \to p'(x)$ be differentiation

   $P_2(\mathbb{R}) \to P_1(\mathbb{R}) \iff a + bx + cx^2 \to b + 2cx \iff \begin{bmatrix} a \\ b \\ c \end{bmatrix} \iff \begin{bmatrix} 0 \\ b \\ 2c \end{bmatrix}$

   Educated guess: nullity $D = 1, \text{rank } D = (1 + n) - 1 = n$

   Proof

   $\text{Ker } D = \{p(x) \in P_n(\mathbb{F}) : D\big(p(x) = \vec{0}\big\} = \{a_1 + a_2 x + ... + a_n x^n\} = \{0 + 0x + ... + 0x^n\} \quad (a_1 = a_2 = ... =$

   $= \{a_1 + 0x^2 + ... + 0x^n : a_i \in \mathbb{F}\} = \text{span } \{1\} \in P_n(\mathbb{F})$

   Since $\{1\}$ is a basis for Ker $D \implies \dim(\text{Ker } D) = 1$
   $\text{Range}(D) = \{q(x) \in P_{n-1}(\mathbb{R}) : q(x) = D(p(x)), \text{for some } p(x) \in P_n(\mathbb{R})\}$

   Claim Range D $= P_{n-1}(\mathbb{R})$
   Range D $\subseteq P_{n-1}(\mathbb{R})$ is trivial

   $P_{n-1}(\mathbb{R}) \subseteq$ Range D
   Let $q(x) = a_0 + a_1 x + ... + a_{n-1} x^{n-1}$
   Then $q(x) = D\big(a_0 + \frac{a_1}{2} x + ... + \frac{a_{n-1}}{n} x^{n-1}\big) \in P_n(\mathbb{R})$
   $q(x) \in \text{Range}(D) \implies \text{rank } L = \dim \text{ Range } L = \dim\ P_{n-1}(\mathbb{R}) = n$

   □

Proof

Rank-Nullity Theorem

Let $B = \{v_1, ..., v_k\}$ be a basis for Ker L, extend $B.$ to obtain $C = \{v_1, ..., v_k, v_{k+1}, ..., v_n\}$

Claim $D = \{L(v_{k+1}), ..., L(v_n)\}$ is a basis for Range L, meaning that

rank $L = \dim$ Range $L = |D| = n - k = \dim V - \text{nullity} L$

Consider $c_{k+1}L(v_{k+1}) + ... + c_k L(v_n) = \vec{0}, \quad \forall c_i \in \mathbb{F}$

$$\begin{aligned}L(c_{k+1}v_{k+1} + ... + c_n v_n) = \vec{0} \implies c_{k+1}v_{k+1} + ... + c_n v_n &\in \text{Ker } L \\ = c_{k+1}v_{k+1} + ... + c_n v_n &= a_1 v_1 + ... + a_k v_k \\ = c_{k+1}v_{k+1} + ... + c_n v_n - a_1 v_1 - ... - a_k v_k &= \vec{0}\end{aligned}$$

Since C is linearly independent, thus $c_i = 0$

We need to prove D spans Range L : span$D$ = Range$L$ to show the claim is true

span$D \subseteq$ Range$L$ is trivial

Range$L \subseteq$ span$D$

Given $w \in$ Range L, show $w \in$ span D.

We can write $a_1v_1 + ... + a_kv_k + a_{k+1}v_{k+1} + ... + a_nv_n$

So $w = L(v) = L(\sum a_iv_i) = a_{k+1}v_{k+1} + ... + a_nv_n \in$ span D

□

> 📘 **Definition — Isomorphism**
> Let $L : V \to W$ be linear mapping
>
> 1. L is **injective**(one-to-one) $\iff$ Ker$L = \{0\} \iff$ nullity$L = 0$
> 2. L is **surjective**(onto) $\iff$ Range$L = W \iff$ rank $L =$ dim$W$
> 3. L is an **isomorphism** if and only if Ker$L = \{0\}$ and Range $L = W$, that is both injective and surjective

Example:

1. Let B be a basis for V, then the coordinate map $[]_B : V \to \mathbb{F}^n$ is an isomorphism

   $\because$ Ker$[]_B = \{0\}$ thus is injective, Range$[]_B = \mathbb{F}^n$ thus is surjective
2. $P_2(\mathbb{R}) \to \mathbb{R}^3$ is an isomorphism
3. $D : P_n(\mathbb{R}) \to P_{n-1}(\mathbb{R}), p(x) \to p'(x)$ be differentiation

   nullity $D = 1 \neq 0$, thus not injective, rank$D =$ dimP_{n-1} is surjective
4. $L : \mathbb{F}^2 \to M_{2\times2}(\mathbb{F}), \begin{bmatrix} a \\ b \end{bmatrix} \to \begin{bmatrix} a+b & a \\ b & a-b \end{bmatrix}$

   We suspect that L cannot be surjective

   Range $L = \left\{ \begin{bmatrix} a+b & a \\ b & a-b \end{bmatrix} : a, b \in \mathbb{F} \right\} = \left\{ a\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + b\begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} : a, b \in \mathbb{F} \right\} =$ span $\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \right\}$

   rank$L \neq$ dim Range L, so L is not surjective

   By rank-nullity theorem, nullity $L =$ dim$\mathbb{F}^2 -$ rank $L = 0$

Given $L : V \to W$ a linear map, we can build a matrix for it as:

1. Choose basis $B = \{v_i, ..., v_n\}$ for V
2. Choose basis $C = \{w_i, ..., w_m\}$ for W
3. Let ${}_C[L]_B = \left[[L(v_1)]_C \ [L(v_2)]_C ... [L(v_n)] \right]_C$

Definition — Proposition

$$L(v) = w \iff [L(v)]_C = [w]_C \iff [L]_B[v]_B = [w]_C$$

That is $[L(v)]_C = {}_C[L]_B[v]_B$

Example:

1. $L : P_2(\mathbb{R}) \to M_{2\times 2}(\mathbb{R}), a + bx + cx^2 \to \begin{bmatrix} c & b \\ a & 0 \end{bmatrix}$

   $B = \{1, x, x^2\}, C = \{E_{11}, E_{12}, E_{21}, E_{22}\}$

   We can view it as a linear map $T : \mathbb{R}^3 \to \mathbb{R}^4, \vec{x} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \to T(\vec{x}) = A\vec{x} = \begin{bmatrix} c \\ b \\ a \\ 0 \end{bmatrix}$

   $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $T(\vec{x}) = A\vec{x}$

2. $L : P_2(\mathbb{F}) \to P_2(\mathbb{F}), L(p(x)) = p(x+1)$. Find ${}_B[L]_B$ where $B = \{1, x, x^2\}$

   $[L]_B = [[L(1)]_B \ [L(x)]_B \ \left[L(x^2)\right]_B] = \left[[1]_B \ [x+1]_B \ \left[x^2 + 2x + 1\right]_B\right] = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

   Use the above matrix to show $L(3 + x) = 4 + x$

   $[L(3+x)]_B = [L]_B[3+x]_B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$

   $L(3 + x) = 4 \cdot 1 + 1 \cdot x + 0 \cdot x^2 = 4 + x$

Definition — Proposition

Let $L : V \to W$ be linear map, let $A = {}_C[L]_B$ where B is basis for V and C be basis for W

1. $v \in \text{Ker } L \iff [v]_B \in \text{Null}(A)$
2. $w \in \text{Range } L \iff [w]_C \in \text{Col}(A)$
3. $v_1, ..., v_k$ is a basis for Ker L if and only if $[v_1]_B, ..., [v_k]_B$ is a basis for $\text{Null}(A)$
4. $w_1, ..., w_l$ is a basis for Range L if and only if $[w_1]_C, ..., [w_l]_C$ is a basis for $\text{Col}(A)$
5. nullity L = nullity A = dim null A
6. rank L = rank A = number of pivots of $\text{RREF}(A)$

Warn: this proposition DID NOT claim Ker $L = \text{Null}(A)$, Range $L = \text{Col}(A)$

Proof

1.

$$\begin{aligned} v \in \text{Ker } L &\iff L(v) = 0 \\ &\iff [L(v)]_C = [0]_C \qquad \text{(by Uniquness of Coordinates)} \\ &\iff {}_C[L]_B[v]_B = [0]_C \\ &\iff A[v]_B = \vec{0} \iff w \in \text{Null}(A) \end{aligned}$$

4. $w_1, ..., w_l$ is a basis for Range L if and only if $[w_1]_C, ..., [w_l]_C$ is a basis for $\text{Col}(A)$

$\Longrightarrow$:

Assume $w_1, ..., w_n$ is abasis for Range L.

Claim: $Z = \left\{[w_1]_C, ..., [w_r]_C\right\}$ is linear independent

Consider $a_1[w_1]_C + ... + a_r[w_r]_C = \vec{0}$

$$[a_1w_1 + ... + a_rw_r] = \begin{bmatrix} 0 \\ ... \\ 0 \end{bmatrix} \Longrightarrow a_1w_1 + ... + a_rw_r = 0$$

By linearity of coordiantes, $\Longrightarrow a_1 = ... = a_r = 0$ since Z is linearly independent

Z spans $\text{Col}(A) = \{A\vec{x} : \vec{x} \in \mathbb{F}^n\}$

$Z \subseteq \text{Col}(A)$ is trivial

$\text{Col}(A) \subseteq Z$, let $A\vec{x} \in \text{Col}(A)$

Note $A\vec{x} = {}_C[L]_B[\vec{v}]_B = [L(\vec{v})]_C$, where $L(\vec{v}) \in \text{Range}(L)$

We can write $A\vec{x} = [L(\vec{v})]_C = [\sum(a_iw_i)]_C = \sum a_i[w_i]_C \in Z$

$\Longleftarrow$: the same argument applies in reverse order.

□

Examples:

1. Consider $L : \mathbb{R}^3 \to P_2(\mathbb{R})$, $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \to (a + 2b - c) + (2a + 2b)x + (3a + 2b + c)x^2$

   Find rank and nullity of L

   $$A = {}_C[L]_B \Longrightarrow A = \left[\left[L\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right)\right]_C \left[L\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right)\right]_C \left[L\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right)\right]_C\right]$$

   $$= \left[[1 + 2x + 3x^2]_C\right][2x + 2x^2]_C[-1 + 2x + x^2]_C] = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

We say V and W are **isomorphic**, denoted as $V \cong W$

Example:

1. If $\dim V = n$, then $V \cong \mathbb{F}^n$

   Pick a basis B for V and consider coordinates map $[]_B : V \to \mathbb{F}^n$ is both injective and surjective, therefore is an isomorphism

2. Even though $P_2(\mathbb{R}) \cong \mathbb{R}^3$

   Not every $L : P_2(\mathbb{R}) \to \mathbb{R}^3$ is isomorphism

   For example, $L(a + bx + cx^2) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is not isomorphism.

3. Let $U \subseteq \mathbb{R}^3$ be a plane through origin, say with equation $x + y + z = 0$

   Then by example 1, $U \cong \mathbb{R}^2$ because $\dim U = 2$.

   An explicit ismorphism : $L : U \to \mathbb{R}^2$

Find a basis for U, $B = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$

Define $L : U \to \mathbb{R}^2$ by $L\left(a \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} a \\ b \end{bmatrix}$

Note we can also write an inverse map: $\begin{bmatrix} a \\ b \end{bmatrix} \mapsto a \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Definition — Linear Map and Isomorphism

1. $L : V \to W$ is an isomorphism if and only if L has an inverse $L^{-1} : W \to V$
2. Let B, C be basis for V and W, Let $A = {}_C[L]_B$, L is isomorphism is and only if A is invertible

Proof

1. L is an isomorphism if and only if L is injective and surjective

if and only if every $w \in W$ where $v \in V$ s.t. $L(w) = W$

Define $L^{-1}V$, check $L^{-1}W = w, L^{-1}V = v$, L^{-1} is linear

2. L is isomorphism if and only if

L^{-1} exists so $L \circ L^{-1} = \text{Id} \implies {}_C\left[L \circ L^{-1}\right]_C = {}_C[\text{Id}]_C = {}_C[L]_B {}_B\left[L^{-1}\right]_C$

Therefore L is isomorphism if and only if

$\exists L^{-1}$ s.t $L \circ L^{-1} = \text{Id} \iff L^{-1}$ s.t. $[L]\left[L^{-1}\right] = I \iff A$ has an inverse

□

Note: ${}_C[L]_B {}_B[T]_D = {}_C[L \circ T]_D$

Example:

Consider $L : P_2\mathbb{R} \to \mathbb{R}^3, a + bx + cx^2 \mapsto \begin{bmatrix} a+b+c \\ b+c \\ c \end{bmatrix}$

Choose $B = \{1, x, x^2\}, C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Then ${}_C[L]_B = \left[[L(1)]_C [L(x)]_C \left[L(x^2)\right]_C \right] = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ which is invertible

Therefore L is an isomorphism

Definition — Isomorphism and Dimensions

Let V, W be vector spaces over the same $\mathbb{F}$. Then $V \cong W \iff \dim V = \dim W$

In fact, bot are isomorphic to $\mathbb{F}^{\dim V}$

Examples:

1. $M_{2\times 2}(\mathbb{R}), P_2(\mathbb{R})$ are not isomorphic because dim $M_{2\times 2}(\mathbb{R}) = 4 \neq 3 =$ dim $P_2(\mathbb{R})$
2. $M_{2\times 2}(\mathbb{R}) \cong \mathbb{R}^4 P \cong P_3(\mathbb{R})$

> ⚠ **Warning** — Isomorphism and Field
>
> $$M_{2\times 2}(\mathbb{R}) \ncong \mathbb{C}^4$$

Change of Coordinates

Let V be a vector space, let $B = \{b_1, ..., b_n\}, C = \{c_1, ..., c_n\}$ be basis for V, let $v \in V$

How are $[v]_B, [v]_C$ related?

$\mathbb{F}^n \leftarrow V \longrightarrow \mathbb{F}^n$

$[v]_C \leftarrow | \ V \mapsto [v]_B$

The composition (take B-coordinates) $\circ$ (take C-coordinates) is a linear map from $\mathbb{F}^n \to \mathbb{F}^n$.
Therefore it has a standard $n \times n$ matrix

$${}_B I_C = [L(e_1)\ L(e_2)\ ...\ L(e_n)] = \left[[c_1]_B\ [c_2]_B\ ...[c_n]_B\right]$$

> 📘 **Definition** — Coordinate Matrix
>
> If $B = \{b_1, ..., b_n\}, C = \{c_1, ..., c_n\}$ be basis for V
>
> We define the change from C coordinates to B-coordinates matrix to be
>
> ${}_B I_C = \left[[c_1]_B\ [c_2]_B\ ...[c_n]_B\right]$

Example:

$S = \{1, x, x^2\}, B = \{2 + 2x - x^2, -2 + 3x + 2x^2, -1 + x + x^2\}$ be bases for $P_2\mathbb{F}$

Then ${}_C I_B = \left[[b_1]_S [b_2]_S ... [b_3]_S\right]$

and ${}_B I_C = ({}_C I_B)^{-1}$